

REVIEW

Wiring the cell signaling circuitry by the NF- κ B and JNK1 crosstalk and its applications in human diseases

J Liu and A Lin

Ben May Department for Cancer Research, The University of Chicago, Chicago, IL, USA

Integration of the cell signaling circuitry determines the ultimate response of a cell to extracellular stimuli. The transcription factor nuclear factor-kappa B (NF- κ B) and mitogen-activated protein kinase JNK1 are major players in the cell signaling circuitry, regulating numerous cellular events and being implicated in the process of many human diseases and certain types of cancer. The interplay between NF- κ B and JNK1 provides a paradigm that shows how the crosstalk between different signaling pathways decides the function of the cell signaling circuitry. Understanding the wiring of the cell signaling circuitry may hold the key for cell signaling-based therapy of human diseases.

Oncogene (2007) **26**, 3267–3278. doi:10.1038/sj.onc.1210417

Keywords: NF- κ B; JNK1; TNF- α ; UV; signaling; cross-talk

Introduction

After decades of intensive research, a blueprint of the cell signaling circuitry is gradually emerging. Many different signaling pathways are integrated into such circuitry to mediate cell proliferation, differentiation and apoptosis. It has been recognized that such integrated signaling network, rather than an individual signaling pathway, determines the ultimate response of a cell to extracellular stimuli. Disruption in such signaling circuitry often leads to cellular pathology including cancer. The transcription factor nuclear factor-kappa B (NF- κ B) and mitogen-activated protein (MAP) kinase (MAPK) JNK1 are two key regulators in one such circuitry, whose deregulation has been implicated in many human diseases including certain types of cancer (Lin and Karin, 2003; Courtois and Gilmore, 2006; Karin, 2006; Lin, 2006).

The regulations of the JNK as well as NF- κ B signaling pathways and their functional consequences have been the subject of several recent reviews (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002; Karin and Lin, 2002; Lin, 2003). This review specifically focuses on defining the integration between NF- κ B- and JNK1-signaling pathways, and how such integration is wired into an apoptotic cell signaling circuitry that can be induced by the proinflammatory cytokine tumor necrosis factor (TNF)- α or ultraviolet irradiation (UV). In addition, the therapeutic potentials of the signaling nodes unraveled in NF- κ B-JNK1 signaling circuitry for the treatment of human diseases including cancer will be discussed.

The JNK signaling pathway

JNK was originally discovered as a protein kinase that can specifically phosphorylate the N-terminal transactivation domain of the transcription factor c-Jun at Ser63 and Ser73 (Hibi *et al.*, 1993). These phosphorylations enhance the transcription activity as well as the transformation ability of c-Jun (Binetruy *et al.*, 1991; Pulverer *et al.*, 1991; Smeal *et al.*, 1991; Adler *et al.*, 1992). Subsequent studies revealed that JNK-family of kinases, belonging to the superfamily of the MAPK consists of three isoforms JNK1, JNK2 and JNK3 (Derijard *et al.*, 1994; Karin, 1995; Davis, 2000). JNK1 and 2 are ubiquitously expressed, whereas JNK3 shows a tissue-specific expression profile (Karin, 1995; Davis, 2000). Many different splice variants are generated from these three isoforms. Recent studies show that most JNK inducers, if not all, significantly activate JNK1, but not JNK2, to exert their biological functions (Liu *et al.*, 2004). This is likely due to the existence of endogenous JNK2 inhibitor(s) that specifically blocks the enzymatic activity of JNK2 (unpublished data). The activation of JNK by extracellular stimuli, from proinflammatory cytokines to physical stresses, is mediated by the MAP kinase module (Lin, 2006), that is, MAP3K $\rightarrow$ MAP2K $\rightarrow$ MAPK, via sequential protein phosphorylation (Figure 1). Phosphorylation of JNK on the T-loop at Thr183 and Tyr185 by two dual specificity MAP2Ks (JNKK1/MKK4/SEK1 and JNKK2/MKK7) is required for its full activation by various extracellular stimuli (Sanchez *et al.*, 1994; Derijard *et al.*, 1995; Lin *et al.*,

Correspondence: Professor A Lin, Ben May Department for Cancer Research, The University of Chicago, 929 East 57th Street, Chicago, IL 60637, USA.

E-mail: alin@huggins.bsd.uchicago.edu

Anning Lin would like to dedicate this review in memory of his father, Jun Lin, who passed away due to heart attack in Nanjing, China, on 28 November 2006.

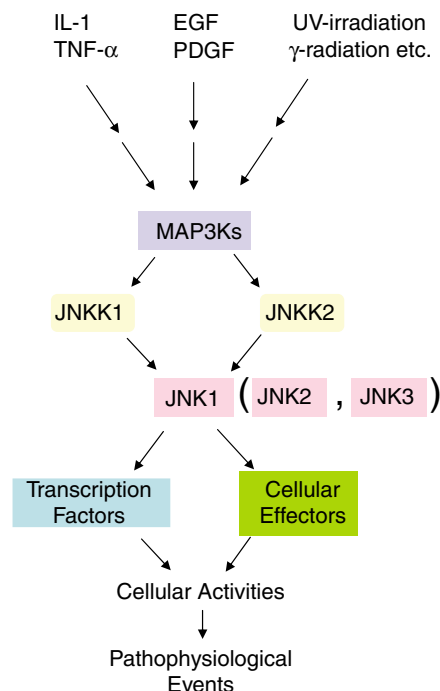


Figure 1 The JNK signaling pathway. JNK is activated by a variety of extracellular stimuli through the MAPK module by sequential phosphorylation, leading to changes in cellular metabolism and gene expression. The JNK signaling pathway regulates numerous cellular activities and is involved in many pathophysiological events.

1995; Lu *et al.*, 1997; Tournier *et al.*, 1997). Several MAP3Ks have been reported to be involved in JNK activation. The list includes members of the mammalian mitogen-activated protein kinase kinase kinase family (Yujiri *et al.*, 1998; Xia *et al.*, 2000), apoptosis signal-regulating kinase 1 (Ichijo *et al.*, 1997), mixed lineage kinase (Tibbles *et al.*, 1996), transforming growth factor-beta (TGF- β)-activated kinase 1 (TAK1) (Shibuya *et al.*, 1996), TPL-2 (Salmeron *et al.*, 1996), and Tao 1 (Hutchison *et al.*, 1998). In addition to c-Jun, JNK phosphorylates and thereby regulates the activities of other transcription factors such as ATF2, Elk-1, p53 and c-Myc. Furthermore, JNK is involved in the regulation of non-transcription factors such as the members of the Bcl-2 family (Bcl-2, Bcl-x_L, Bim, Bmf and Bad) (Davis, 2000; Chang and Karin, 2001; Lin, 2006). Biochemical characterization and genetic analysis reveal that JNK plays a central role in regulation of many cellular activities, such as proliferation, differentiation, migration, transformation and programmed cell death (apoptosis and necrosis; cell death for the simplicity hereinafter) (Lin, 2006). Deregulation of the JNK activity has been implicated in many human diseases including certain types of cancer (Antonyak *et al.*, 2002; Cripe *et al.*, 2002; She *et al.*, 2002), cardiac hypertrophy and ischemia (He *et al.*, 1999; Nemoto *et al.*, 1998; Sadoshima *et al.*, 2002), immune disorders (Reimold, 2002), liver injury (Uehara *et al.*, 2005), obesity (Hirosumi *et al.*, 2002) and neurodegenerative diseases like Alzheimer's and Parkinson's disease

(Xia *et al.*, 2001; Okazawa and Estus, 2002). Not surprisingly, JNK activation is tightly regulated by many cellular regulators or modulators, such as scaffold proteins (Whitmarsh *et al.*, 1998; Ito *et al.*, 1999; McDonald *et al.*, 2000) and MAP phosphatases (Karin, 1995). Recent studies have revealed that the JNK activity can be negatively or positively regulated by NF- κ B for inhibition or promotion of cell death induced by TNF- α and UV, respectively (Tang *et al.*, 2001; Liu *et al.*, 2006).

The NF- κ B signaling pathway

The heterodimeric transcription factor NF- κ B regulates immune responses, inflammation, infections, oncogenesis and cell death (Siebenlist *et al.*, 1994; Thanos and Maniatis, 1995; Verma *et al.*, 1995; Baeuerle and Baltimore, 1996; Baldwin, 1996; Barnes and Karin, 1997; Ghosh *et al.*, 1998; Karin and Lin, 2002). Deregulation of NF- κ B activity has been implicated in numerous human diseases including cancer (Baldwin, 1996; Karin and Lin, 2002). There are five mammalian Rel/NF- κ B proteins: RelA(p65), c-Rel, RelB, NF- κ B1 (p50/p105) and NF- κ B2 (p52/p100) (Baldwin, 1996). Among them, p50 and p52 are the proteolytic products of their precursor p105 and p100, respectively (Baldwin, 1996). NF- κ B is a latent transcription factor. In non-stimulated cells, NF- κ B proteins are mainly retained in the cytoplasm through their interaction with the cytoplasmic inhibitors. RelA and c-Rel interacts with inhibitors of NF- κ B (I κ Bs), whereas RelB with NF- κ B2p100, which has an I κ B-like C-terminus (Baldwin, 1996). A key step in regulation of NF- κ B activity by extracellular stimuli is to control its translocation from the cytoplasm to the nucleus (Baldwin, 1996). This process is regulated by at least three major signaling pathways as shown in Figure 2, namely (1) the I κ B kinase (IKK)-dependent canonical; (2) IKK-dependent non-canonical (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002); and (3) IKK-independent the p38-CK2 pathways (Kato *et al.*, 2003). In response to a variety of extracellular stimuli such as TNF- α , interleukin (IL)-1 β and lipopolysaccharide (LPS), the NF- κ B dimers containing RelA or c-Rel are activated via the canonical pathway, in which I κ B proteins are phosphorylated by the I κ B kinase complex (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002) thereby freeing the NF- κ B dimers. The IKK complex has two catalytic subunits, IKK α and IKK β , and two regulatory subunits IKK γ /NEMO and ELKS (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002; Ducut Sigala *et al.*, 2004). IKK β is essential for NF- κ B activation in the canonical pathway (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002). Phosphorylation of I κ B α by IKK triggers the ubiquitination and subsequent proteasome-mediated degradation of I κ B α , thereby freeing the NF- κ B dimers containing RelA or c-Rel (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002). Stimulation by lymphotoxin β and other signals that are involved in adaptive immunity results in activation of the NF- κ B dimers containing

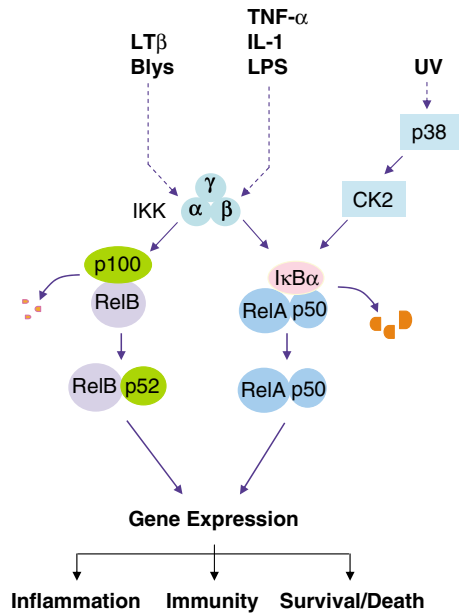


Figure 2 The NF- κ B signaling pathway. NF- κ B is activated by IKK-dependent canonical and non-canonical pathways as well as IKK-independent p38-CK2 pathways. Although IKK β is required for the canonical pathway activated by stimuli like TNF- α , IL-1 β and LPS, IKK α is involved in the non-canonical pathway triggered by stimuli such as lymphotoxin (LT- β) and Blys (BAFF). The p38-CK2 pathway is activated by UV. All three pathways lead to the ubiquitination of I κ B or p100, followed by proteasome-mediated proteolysis. The free NF- κ B translocates into the nucleus, where it stimulates expression of numerous genes that are involved in many cellular activities.

RelB and p52 through the non-canonical pathway, in which NF- κ B2p100 proteins are phosphorylated by the IKK complex containing IKK α homodimers (Karin and Lin, 2002). This phosphorylation event targets NF- κ B2p100 for ubiquitination and proteasome-mediated degradation, thereby releasing the RelB:p52 dimers. Activation of NF- κ B by UV and the hepatitis B virus trans-acting factor pX is independent of the IKK complex (Bender *et al.*, 1998; Li and Karin, 1998; Purcell *et al.*, 2001). In the case of UV, I κ B α proteins are phosphorylated by CK2, followed by ubiquitination and degradation (Kato *et al.*, 2003). Mechanisms of activation apart, the free NF- κ B dimer translocates into the nucleus, where it stimulates expression of genes that are involved in many cellular activities. Phosphorylation of the Rel proteins in the nucleus, which is most likely mediated by the phosphatidyl inositol 3 kinase (PI3K)-Akt and p38 MAPK pathways, is required for full induction of NF- κ B target gene (Madrid *et al.*, 2000; Zhang *et al.*, 2006). Although the nuclear translocation of the majority NF- κ B proteins is signal-dependent, emerging evidence shows that a small portion of NF- κ B proteins resides even in the nucleus of non-stimulated cells. This is due to the low level, but constitutive degradation of I κ B proteins, a process that is controlled by CK2-mediated phosphorylation of the C-terminus of I κ B α proteins (Pando and Verma, 2000). Accumulating evidence shows that NF- κ B is a key regulator of the cell

signaling circuitry (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002), either negatively or positively regulating the activity of other signaling pathways, including the JNK pathway (Tang *et al.*, 2001; Liu *et al.*, 2006), in response to various extracellular stimuli.

Wiring the TNF- α signaling circuitry for cell survival by the negative crosstalk between NF- κ B and JNK1

The TNF- α signaling circuitry and cell death

TNF- α is a pleiotropic proinflammatory cytokine, which regulates many pathophysiological activities, including immune responses, inflammation, proliferation and cell death. TNF- α exerts its biological functions through the activation of multiple downstream effectors, including NF- κ B, JNK and caspase (Karin and Lin, 2002) (Figure 3). TNF- α activates NF- κ B and JNK through the TNF-receptor complex I (TNF-R1), which contains the trimeric TNF-R1, TNF-R1-associated death domain protein, TNF-R1-associated factor 2 (TRAF2) and receptor interaction protein 1 (Micheau and Tschopp, 2003). Whereas TNF- α -induced cell death is mediated by the TNF-R1 complex II, which is composed of the internalized complex I, Fas-associated death domain protein, and the initiation caspases, caspase 8 and caspase 10 (Micheau and Tschopp, 2003). However, TNF- α does not typically induce cell death unless the activation of NF- κ B is blocked (Karin and Lin, 2002). This is because some of NF- κ B target genes encode inhibitors of apoptosis, such as c-IAP1, c-IAP2, XIAP and c-FILP, which inhibit caspase activity (Karin and Lin, 2002). The prosurvival role of the IKK-NF- κ B pathway in TNF- α -induced cell death is further supported by the observations that genetic disruption of RelA, which is the major transactivating subunit of NF- κ B (Berg and Baltimore), or IKK β , which is essential for NF- κ B activation by TNF- α (Li *et al.*, 1999a, b; Tanaka *et al.*, 1999), results in TNF- α -induced massive hepatocyte death in mouse liver. By contrast, the role of JNK in TNF- α -induced apoptosis has been highly controversial (Lin, 2003). Biochemical studies show that JNK can be prodeath, antideath or has no role in the cell death process (Liu *et al.*, 1996; Verheij *et al.*, 1996; Natoli *et al.*, 1997).

The crosstalk between NF- κ B and JNK determines TNF- α cytotoxicity

Recent studies have uncovered that the role of JNK in TNF- α -induced cell death depends on the activity of the IKK-NF- κ B pathway (De Smaele *et al.*, 2001; Tang *et al.*, 2001). In wildtype (wt) mouse embryonic fibroblasts (MEFs) and many other cell types, TNF- α induced transient JNK activation, which reached its maximal activity at 15 min and returned to near basal at 30 min (Tang *et al.*, 2001). By contrast, TNF- α induced prolonged JNK activation in MEFs deficient in IKK β (Tang *et al.*, 2001) or RelA (De Smaele *et al.*, 2001; Tang *et al.*, 2001), or in cells expressing a dominant negative I κ B α mutant (De Smaele *et al.*, 2001). More importantly, conversion of JNK activation from prolonged to

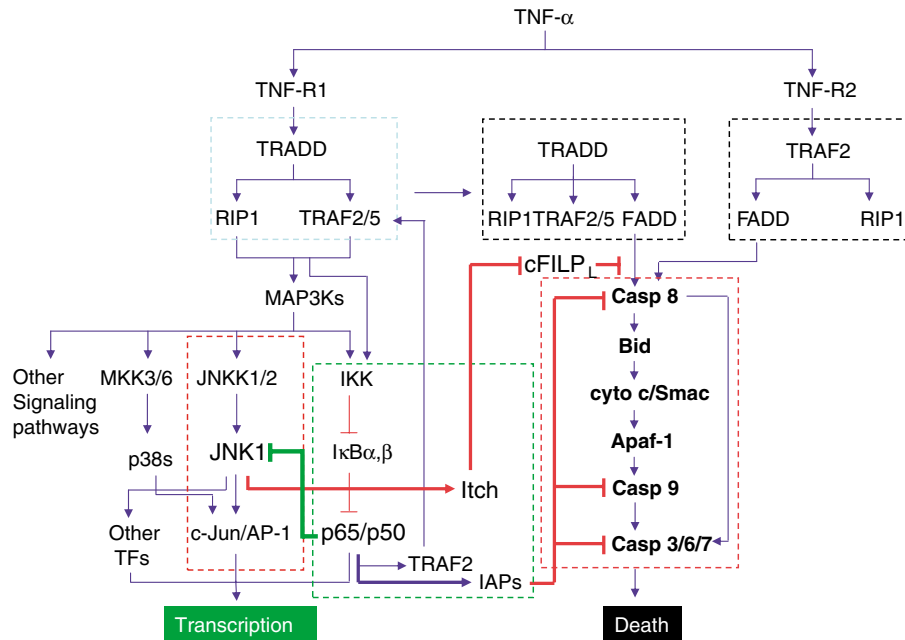


Figure 3 The crosstalk between NF- κ B and JNK1 wires the TNF- α signaling circuitry for cell survival. In addition to the inhibition of caspases, NF- κ B prevents TNF- α -induced prolonged JNK1 activation. Prolonged JNK1 activation results in phosphorylation and activation of the E3 ligase Itch, which catalyses the ubiquitination of the caspase 8 inhibitor c-FLIP_L and triggers its proteolysis by the proteasome. This breaks the brake on TNF- α -activated cell death pathway, leading to cell death.

transient, blocked TNF- α -induced apoptosis in the absence of NF- κ B activation (Tang *et al.*, 2002). These findings demonstrated that NF- κ B prevents TNF- α -induced prolonged JNK activation, which otherwise contributes to TNF- α -induced cell death (Lin, 2003). It is important to point out that NF- κ B does not selectively inhibit prolonged JNK activation. Activation of NF- κ B by TNF- α takes about 15 min and the JNK-inhibition by NF- κ B also depends on *de novo* protein synthesis (Tang *et al.*, 2001). Thus, NF- κ B can only inhibit the concurrently TNF- α -activated JNK after NF- κ B itself has been activated. In support of this notion, preactivation of NF- κ B did block TNF- α -induced transient- as well as prolonged- activation of JNK (J Liu and A Lin, unpublished data). Thus, NF- κ B inhibits TNF- α -induced apoptosis through at least two mechanisms: inhibition of caspase activation and prevention of prolonged JNK activation (Lin, 2003). The inhibition of TNF- α -induced prolonged JNK activation by NF- κ B is also critical to suppress TNF- α cytotoxicity *in vivo*. Although TNF- α induced prolonged JNK1 activation in *Ikk β ^{-/-}* mice and massive death of hepatocytes resulting in embryonic lethality, the cytotoxicity of TNF- α was significantly delayed in *Ikk β ^{-/-} Jnk1^{-/-}* mice (Chang *et al.*, 2006). This is also the case with liver injury in adult mice. Administration of concanavalin A (ConA) into hepatocyte-specific *Ikk β* knockout mice induced prolonged JNK activation and liver degeneration, whereas ConA cytotoxicity was blocked in JNK1-deficient mice (Maeda *et al.*, 2003). Similarly, genetic inactivation of *Jnk1* alleles in mice reduced liver cytotoxicity of diethylnitrosamine, a chemical carcinogen that activates JNK, in hepatocyte-specific

Ikk β knockout mice (Maeda *et al.*, 2005). Between two ubiquitously expressed JNK isoforms, JNK1 but not JNK2 is significantly activated by TNF- α and is required for TNF- α -induced cell death (Liu *et al.*, 2004), whereas JNK2 is involved in negative regulation of c-Jun (Liu *et al.*, 2004; Sabapathy *et al.*, 2004). It is important to note that the inhibition of prolonged JNK activation by NF- κ B is highly specific, as the kinetics of JNK activation by other extracellular stimuli such as IL-1 and UV that can activate both JNK and NF- κ B was not affected by loss of NF- κ B activity (Tang *et al.*, 2001). It is possible that NF- κ B-induced inhibitor(s) may selectively inhibit a unique signaling component(s) in the TNF- α signaling pathway that leads to JNK activation, or eliminate a positive regulator of JNK activation.

NF- κ B inhibits TNF- α -induced prolonged JNK activation through multiple mechanisms

Several scenarios have been proposed to explain how NF- κ B inhibits TNF- α -induced prolonged JNK activation. The first scenario is that NF- κ B may induce specific JNK inhibitors. Ectopic expression of X-chromosome-linked IAP (XIAP), which is an NF- κ B target gene (Tang *et al.*, 2001), blocked TNF- α -induced prolonged JNK activation in the absence of NF- κ B activation. Yet, the kinetics of TNF- α -induced JNK activation was not altered in *Xiap* null MEFs (G Tang and A Lin, unpublished data). It is possible that the inhibitory effect of XIAP on JNK activation may be compensated by other NF- κ B-induced inhibitors, such as Gadd45 β (De Smaele *et al.*, 2001). Gadd45 β was identified as an NF- κ B-induced JNK inhibitor by the virtue of its

ability to inhibit prolonged JNK activation and cell death in TNF- α -treated RelA null MEFs (De Smaele *et al.*, 2001). In addition, Gadd45 β was reported to specifically bind to and inhibit JNKK2/MKK7, thereby preventing TNF- α -induced prolonged JNK activation (Papa *et al.*, 2004). It has been postulated that Gadd45 β may be a cell-type-specific JNK inhibitor, but the underlying mechanism by which Gadd45 binds to JNKK2/MKK7 in a cell-type-dependent manner has yet to be demonstrated (Papa *et al.*, 2004). However, inactivation of Gadd45 also failed to affect the kinetics of TNF- α -induced JNK activation in MEFs (Amanullah *et al.*, 2003). This is quite surprising in light of the putative inhibitory role of Gadd45.

Another scenario is that activation of NF- κ B results in the elimination of reactive oxygen species (ROS), whose accumulation otherwise leads to TNF- α -induced prolonged JNK activation (Sakon *et al.*, 2003). In *Traf2*^{-/-} *Traf5*^{-/-} MEFs, TNF- α was unable to activate NF- κ B, but still induced cell death, which was accompanied by accumulation of ROS and delayed but sustained JNK phosphorylation (Sakon *et al.*, 2003). Interestingly, TNF- α -induced sustained JNK phosphorylation and cell death were blocked by antioxidants like butylated hydroxyanisole, which is a ROS inhibitor (Sakon *et al.*, 2003). Thus, it has been proposed that ROS accumulation may be responsible for TNF- α -induced prolonged JNK activation in the absence of NF- κ B (Sakon *et al.*, 2003). This notion is supported by a recent report showing that ROS inactivated several mitogen-activated protein phosphatases (MKPs), thereby contributing to TNF- α -induced prolonged JNK activation in the absence of NF- κ B activation (Kamata *et al.*, 2005). It has also been reported that ferritin heavy chain (FHC), whose expression is regulated by NF- κ B, inhibited TNF- α -induced ROS accumulation, prolonged JNK activation and cell death when it was overexpressed in RelA null MEFs (Pham *et al.*, 2004). Whether FHC is a physiological NF- κ B-induced JNK inhibitor has yet to be demonstrated. Taken together, the above findings suggest that NF- κ B may upregulate expression of FHC, which in turn inhibits ROS accumulation; this releases the inhibition on MKPs, thereby blocks TNF- α -induced prolonged JNK activation.

Although ROS accumulation is likely to play an important role in TNF- α -induced prolonged JNK activation and cell death, it may not be the sole mechanism for TNF- α -induced prolonged JNK activation when NF- κ B activation is blocked. Although TNF- α -induced JNK activation in RelA^{-/-} MEFs becomes prolonged as early as 30 min after TNF- α stimulation (Tang *et al.*, 2001), the kinetics of the RelA-FHC-MPK axis is much slower in TNF- α -stimulated cells: the induction of FHC by NF- κ B occurred at 1–2 h (Pham *et al.*, 2004), the earliest detectable ROS accumulation was at 1–2 h (Sakon *et al.*, 2003) and the inactivation of MKPs by ROS was detected several hours later (Kamata *et al.*, 2005). It is also not clear whether the NF- κ B-FHC-MKP axis has any inhibitory effect on JNKK1/MKK4 and JNKK2/MKK7, whose activation

is prolonged in TNF- α -treated RelA null MEFs as well (Lin, 2003; J Liu, G Tang, Y Minemoto and A Lin, unpublished data). It is plausible that ROS accumulation may account for the late phase (> 2 h) of prolonged JNK activation in TNF- α -treated RelA null MEF, whereas the early phase of prolonged JNK activation (< 2 h) may be regulated by other mechanisms.

NF- κ B may prevent TNF- α -induced prolonged JNK activation by negatively regulating the activation of JNK upstream activators. In *Drosophila*, the RelA homologous *Relish* was found to inhibit LPS-induced sustained JNK activation by promoting proteasome-dependent degradation of the MAP3K TAK1 (Park *et al.*, 2004), although the underlying mechanism has yet to be determined. Whether TAK1 is targeted by NF- κ B-mediated inhibition in TNF- α -treated mammalian cells is unknown. Another upstream activator of JNK that might be targeted by NF- κ B-mediated inhibition is JNKK2/MKK7. It is known that TNF- α not only induces prolonged activation of JNK, but also sustained activation of its upstream kinases, JNKK1/MKK4 and JNKK2/MKK7 in RelA null MEFs, as discussed above (Lin, 2003; J Liu, G Tang, Y Minemoto and A Lin, unpublished data). It has been reported that TNF- α -induced JNKK2/MKK7 activation was selectively inhibited by Gadd45 β through direct protein–protein interaction (Papa *et al.*, 2004). However, both TNF- α and IL-1 β activate NF- κ B and also use JNKK2/MKK7 to activate JNK (Lin, 2003). If the inhibition of TNF- α -activated prolonged JNK activation by NF- κ B is indeed mediated by direct binding of Gadd45 β to JNKK2/MKK7 (Papa *et al.*, 2004), an explanation is needed about how NF- κ B does not inhibit IL-1 β -induced JNK activation. Whether TNF- α induces prolonged activation of JNKK2/MKK7 in Gadd45 β null cells has also yet to be determined. As TNF- α -induced activation of JNKK1/MKK4 was also prolonged (Lin, 2003; J Liu, G Tang, Y Minemoto and A Lin, unpublished data), it is not clear how inhibition of JNKK2/MKK7 by Gadd45 β alone could prevent prolonged JNK activation. Future studies are needed to determine whether other JNK upstream activators are negatively regulated by NF- κ B, and if so, what the regulatory mechanism is. Taken together, it appears that multiple mechanisms are involved in NF- κ B-mediated inhibition of TNF- α -induced prolonged JNK activation.

Prolonged activation of JNK1 contributes to TNF- α -induced cell death through elimination of caspase inhibitor(s)

Another remaining question is how the prolonged activation of JNK contributes to TNF- α -induced apoptosis in the absence of NF- κ B activation. JNK is not a genetic component in the cell death pathway or a 'proapoptotic' protein kinase (Lin, 2003). Although JNK activation is involved in cell death induced by many death insults, not all JNK activators induce cell death. More importantly, prolonged JNK activation is only able to promote, but not sufficient to induce cell death even in the absence of NF- κ B activation (Lin, 2003; Chang *et al.*, 2006; G Tang, Y Minemoto and A

Lin, unpublished data). Ectopic expression of the constitutively active JNK (the JNKK2-JNK1 fusion protein) (Zheng *et al.*, 1999), or the dominant-negative mutant of MKP for JNK (Kamata *et al.*, 2005), does not induce cell death even in combination with TNF- α . Furthermore, genetic and biochemical analyses show that JNK has antiapoptotic functions. Activation of JNK by hematopoietic cytokines, such as IL-3, suppresses growth factor withdrawal-induced cell death in hematopoietic cells, through phosphorylation and inactivation of the prodeath Bcl-2 family protein Bad (Yu *et al.*, 2004). JNK also suppresses cell death in certain tumors (Antonyak *et al.*, 2002; Cripe *et al.*, 2002; She *et al.*, 2002). Based on these observations, it has been proposed that JNK regulates cell death by 'breaking the brake on apoptosis' (Lin, 2003). According to this model, JNK activation eliminates suppressors of apoptosis to promote, but does not induce apoptosis (Lin, 2003). In support of this model, prolonged JNK activation was found to be required for the proteolysis of the pro-apoptotic Bcl-2 family protein Bid, generating a novel form of Bid, jBid (Deng *et al.*, 2003). jBid was found to induce selectively the release of second mitochondria-derived activator of caspase (Smac), which in turn inactivated the caspase 8 inhibitor c-IAP1 (Deng *et al.*, 2003). This removes the inhibition on caspase 8 so that it can be activated for cell death. However, whether jBid is a physiologically relevant effector of prolonged JNK activity is debatable, as c-IAP1-deficient mice are not hypersensitive to death insults (Wu *et al.*, 2005). It is also unknown how prolonged JNK activation leads to the proteolysis of Bid. More recently, it has been reported that prolonged JNK1 activation promotes TNF- α -induced cell death through the Itch-cFLIP_L axis (Chang *et al.*, 2006). The E3 ligase Itch is a newly identified substrate of JNK1, but not JNK2 (Gao *et al.*, 2004). Phosphorylation and activation of Itch by JNK1 resulted in ubiquitination of the specific caspase 8 inhibitor cFLIP_L, triggering its degradation by the proteasomes (Chang *et al.*, 2006). It is important to note that although NF- κ B negatively regulates JNK1 activation for cell survival, prolonged JNK1 activation via Itch eliminates cFLIP_L, thereby underlining the protection by NF- κ B. It is proposed that prolonged, but not transient, JNK1 activation is required for sustained activation of Itch so that cFLIP_L can be significantly eliminated by the proteasomes (Chang *et al.*, 2006). Importantly, the JNK-Itch-cFLIP_L axis plays a critical role in TNF- α -induced cytotoxicity *in vivo*, as demonstrated in two different acute liver failure mouse model systems (Chang *et al.*, 2006). Thus, the finding that the prolonged activation of JNK eliminates cFLIP_L via Itch may provide the long-sought missing link between prolonged JNK1 activation and TNF- α -induced cell death. Since elimination of cFLIP_L releases the inhibition on caspase 8, but the release of the inhibition itself is unable to activate the caspase, this finding provides strong evidence that prolonged JNK1 promotes, but does not initiate cell death, as predicted by the 'break the brake on apoptosis' model (Lin, 2003).

Wiring the UV signaling circuitry for cell death by the positive crosstalk between NF- κ B and JNK1

The UV signaling circuitry and cell death

UV is a major environmental stress and a potent carcinogen in skin cancer (Herrlich *et al.*, 1997). UV exerts its biological functions through the activation of multiple downstream signaling pathways, including NF- κ B, JNK and caspases (Bender *et al.*, 1998; Li and Karin, 1998). This results in the induction of many genes, a phenomenon known as the mammalian UV response (Herrlich *et al.*, 1997). UV elicits cell death through both membrane death receptor-dependent extrinsic death pathway and ROS-dependent intrinsic mitochondria-dependent death pathway (Figure 4). UV rapidly induces robust and sustained JNK through the cytoplasmic membrane receptors such as tyrosine kinase receptors and TNF-R1 (Rosette and Karin, 1996), as well as DNA damage (Lee *et al.*, 1998; Khelifi *et al.*, 2005). Recent studies show that JNK1, but not JNK2 is essential for UV-induced cell death (Liu *et al.*, 2004, 2006). UV induces NF- κ B activation through the p38-CK2 signaling pathway, with a slow kinetics (2–4 h) (Bender *et al.*, 1998; Li and Karin, 1998; Kato *et al.*, 2003; Liu *et al.*, 2006). Although NF- κ B is a major survival factor that protects cells from TNF- α and other death signals (Beg and Baltimore, 1996; Liu *et al.*, 1996; Van Antwerp *et al.*, 1996), it promotes UV-induced cell death. Unlike TNF- α , UV-activated NF- κ B does not inhibit UV-induced JNK activation (Liu *et al.*, 2006). In fact, NF- κ B cooperated with the JNK-c-Jun pathway to upregulate the proapoptotic Fas ligand (FasL), which stimulates the extrinsic death pathway through the autocrine mechanism (Kasibhatla *et al.*, 1998). In addition, NF- κ B was found to down-regulate the antiapoptotic Bcl-2 family protein Bcl-x_L in response to UV and other DNA-damage agents (Campbell *et al.*, 2004). However, it was puzzling that how the slow activation of NF- κ B could possibly regulate the rapid onset of UV-induced cell death (Nijhawan *et al.*, 2003).

NF- κ B promotes UV-induced cell death through augmentation of UV-induced rapid and robust JNK activation

Recently, it has been shown that NF- κ B may regulate the onset of UV-induced cell death through the promotion of JNK1 activation (Liu *et al.*, 2006) (Figure 4). UV-induced JNK1 activation as well as UV-induced cell death was significantly reduced, although still sustained, in MEFs deficient in RelA or expressing RelA siRNA (Liu *et al.*, 2006). The reduction of JNK1 activation is responsible for the resistance of RelA null MEFs to UV-induced cell death. UV at 10 J/m² induced strong JNK1 activity and significant cell death in wt fibroblasts, but low JNK1 activity and little cell death in RelA null MEFs (Liu *et al.*, 2006). However, RelA null MEFs treated with 60 J/m² UV had JNK1 activity comparable to that of wt fibroblasts treated with 10 J/m² UV (Liu *et al.*, 2006). Under these conditions, the death rate was similar between wt and

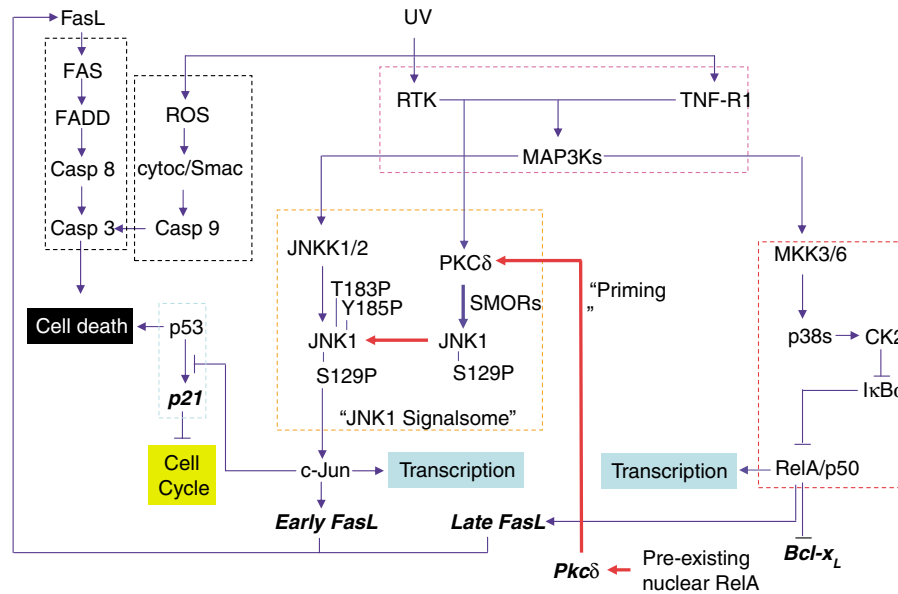


Figure 4 The crosstalk between NF- κ B and JNK1 wires the UV signaling circuitry for cell death. In non-stimulated cells, the pre-existing nuclear RelA/NF- κ B has already been recruited to the promoter of PKC δ and is essential for expression of the latter. When stimulated by UV, PKC δ phosphorylates JNK1 at Ser129 and the phosphorylation is required for further phosphorylation of JNK1 by upstream MAP2Ks so it can be fully activated. Thus, the priming by RelA/NF- κ B via PKC δ ensures UV to rapidly induce robust JNK1 activation, which is essential for UV-induced cell death.

RelA null MEFs (Liu *et al.*, 2006). Consistent with these findings, ectopic expression of a constitutively active JNK (JNKK2-JNK1), sensitized RelA null MEFs for UV-induced cell death (Liu *et al.*, 2006). Thus, the strength of JNK1 activation determines whether it contributes to UV-induced cell death.

The 'priming' effect of the pre-existing nuclear RelA/NF- κ B in UV-induced cell death

UV induces rapid and robust activation of JNK1 (15 min) but slow activation of NF- κ B (2–4 h) (Bender *et al.*, 1998; Li and Karin, 1998; Liu *et al.*, 2006). How could NF- κ B positively regulate UV-induced JNK1 activation? It turned out that the pre-existing nuclear RelA/NF- κ B, rather than activated NF- κ B, is required for UV-induced JNK1 activation (Liu *et al.*, 2006). In non-stimulated cells, RelA had already been recruited to the promoter and is required for expression of protein kinase C δ (PKC δ), which is essential for UV-induced rapid and robust JNK1 activation and cell death (Liu *et al.*, 2006; also see below). The pre-synthesized PKC δ proteins in non-stimulated cells are critical for its rapid activation by UV – a 'priming' effect in UV signaling (Liu *et al.*, 2006) (Figure 4). The pre-existing nuclear RelA was also recruited to the promoter of another NF- κ B target gene Cox-2 for expression (Liu *et al.*, 2006). Like RelA, RelB was found to be recruited to the promoters of PKC δ and Cox-2 (Liu *et al.*, 2006). By contrast, c-Rel and p50 were selectively recruited to the promoters of I κ B α and TRAF2 in non-stimulated cells (Liu *et al.*, 2006). It appears that before stimulation, the pre-existing nuclear NF- κ B subunits are differentially recruited to promoters of many NF- κ B target genes – some of which encode key components of the cell

signaling circuitry – to regulate their expression. Thus, one of the physiological functions of the pre-existing nuclear NF- κ B is to prime cells so that they can respond properly to extracellular stimuli. Future studies are needed to test this 'priming' hypothesis in other physiological or pathological settings.

The RelA-PKC δ axis regulates the assembly of UV-induced JNK1 signalsome

Despite being the most potent activator of JNK1, the signaling pathway that leads to JNK1 activation by UV is incompletely understood. Using pharmacological inhibitors and dominant-negative mutants, a recent report showed that the classic PKC isoforms, such as PKC α and PKC β 2, are required for JNK activation by UV, TNF- α and TPA (Lopez-Bergami *et al.*, 2006). However, genetic analysis revealed that a novel PKC isoform PKC δ is essential for UV-induced rapid and robust JNK1 activation (Liu *et al.*, 2006). PKC δ is a *bona fide* NF- κ B target gene, whose expression depends on the pre-existing nuclear RelA/NF- κ B (Liu *et al.*, 2006). The augmentation by the pre-existing NF- κ B on UV-induced JNK1 activation was abolished by the genetic inactivation of PKC δ alleles, but not by inactivation of PKC α alleles or knockdown of PKC β 2 by small interfering RNA (siRNA) (Liu *et al.*, 2006). Consistent with these findings, biochemical studies have shown that UV activated only PKC δ but not PKC α or PKC β 2 (Liu *et al.*, 2006). Thus, JNK1 activation by UV is selectively regulated by PKC δ , whereas activation of JNK1 by other stimuli like TPA may require PKC α or PKC β 2. The involvement of PKC δ in UV-induced JNK1 activation is highly specific, as PKC δ does not regulate JNK1 activation by other extracellular stimuli

such as TNF- α and IL-1 β , nor UV-induced activation of another MAP kinase p38 (Liu *et al.*, 2006).

The presence of PKC δ proteins before UV stimulation is critical for UV to induce rapid and robust JNK1 activation. Phosphorylation of JNK1 by PKC δ at Ser129 is a prerequisite for the subsequent phosphorylation by the upstream MAP2Ks for its full activation (Liu *et al.*, 2006). Although the mechanism remains unknown, PKC δ -mediated phosphorylation appears to affect the assembly of UV-induced JNK1 signalsome (J Liu and A Lin, unpublished data). Proteomic analysis revealed that several proteins that may act as signal-specific modulators or regulators (SMORs) are recruited to the UV-induced JNK1 signalsome in a PKC δ -dependent manner (Figure 4; J Liu and A Lin, unpublished data). Future studies will determine whether these putative SMORs are involved in the regulation of UV-induced JNK1 activation by the RelA-PKC δ axis.

JNK1 contributes to UV-induced cell death through the promotion of both cytoplasmic and nuclear death events
The prodeath function of JNK1 in UV-induced cell death is mediated by multiple mechanisms. Accumulating evidence indicates that JNK1 is involved in UV-activated intrinsic mitochondria-dependent death pathway. It has been shown that the JNK1 effector c-Jun blocks p53-induced expression of the cell cycle inhibitor p21 (Shaulian *et al.*, 2000). This unleashes p53 to induce cell death through the activation of the prodeath Bcl-2 family protein Bax, which activates the mitochondrial death pathway (Kasibhatla *et al.*, 1998). It has also been proposed that JNK is required for UV-induced cytochrome *c* release. In *Jnk1*^{-/-}*Jnk2*^{-/-} MEFs, UV-induced activation of the intrinsic death pathway was inhibited, including the release of cytochrome *c* from the mitochondria (Tournier *et al.*, 2000). The notion that JNK may regulate the release of cytochrome *c* is further supported by the observation that JNK1 phosphorylated the BH3-only Bcl-2 family proteins Bim and Bmf, both of which are involved in the regulation of the cytochrome *c* release, and promoted their translocation from the cytoplasm to the mitochondria (Lei and Davis, 2003). Yet, how JNK1-mediated phosphorylation regulates the subcellular localization of Bim and Bmf still remains to be elucidated. It also has to be determined whether the JNK-Bim/Bmf axis is essential for JNK1 to contribute UV-induced cell death.

The extrinsic membrane-receptor-dependent death pathway is also a target of JNK1 in UV-induced cell death. Biochemical studies have shown that JNK, via c-Jun, cooperates with activated NF- κ B to upregulate FasL expression in response to UV and other death insults (Kasibhatla *et al.*, 1998). Genetic analysis has further revealed that the rapid and robust activation of JNK1 is essential for the induction of the early-phase expression of FasL by the pre-existing RelA/NF- κ B and not for the suppression of Bcl-x_L by the activated NF- κ B (Liu *et al.*, 2006). The early-phase FasL expression contributes to the onset of UV-induced cell death through the autocrine mechanism activating the extrinsic

death pathway. By contrast, the late-phase expression of FasL, which is regulated by UV-activated NF- κ B, may reinforce the death process through the further activation of extrinsic death pathway (Liu *et al.*, 2006).

In addition to the regulation of cytoplasmic death pathways, JNK1 is also involved in the nuclear death events such as DNA fragmentation (Lu *et al.*, 2006). It has been identified that DNA fragmentation is carried out by CAD or caspase-activated DNase (Enari *et al.*, 1998). Recently, it has been shown that in UV-irradiated cells, JNK1 phosphorylates histone H2AX, a variant of the histone H2A family, and this phosphorylation of H2AX is required for DNA fragmentation by CAD (Lu *et al.*, 2006). However, further studies are required to determine the mechanism by which JNK1-mediated phosphorylation regulates the prodeath activity of H2AX and to evaluate whether JNK1 regulates DNA fragmentation induced by other death signals as well through phosphorylation of H2AX.

Whether c-Jun is involved in the prodeath function of JNK1 in UV-induced cell death has remained controversial. An earlier report showed that UV-induced cell death was enhanced in c-Jun null MEFs, suggesting that unlike JNK, c-Jun may protect cells from UV-induced cell death (Wisdom *et al.*, 1999). This is contradictory to a recent report suggesting that JNK contributes to UV-induced cell death through c-Jun-mediated inhibition on p53-dependent transcription of p21, through which the proapoptotic activity of p53 is unleashed (Shaulian *et al.*, 2000). However, it should be noted here that it is possible that the role of c-Jun in UV-induced cell death depends on the strength of UV signal. At low dose of UV, p53 is capable of inducing cell-cycle arrest through induction of p21 but this induction is blocked by JNK1 via c-Jun (Shaulian *et al.*, 2000). Genetic inactivation of JNK1 or c-Jun allows p53 to induce cell cycle arrest, thereby suppressing UV-induced cell death (Shaulian *et al.*, 2000). By contrast, the ability of p53 to induce cell cycle arrest is impaired at high doses of UV. Under this condition, UV-induced apoptosis will be no longer promoted by c-Jun and JNK1 uses other mechanisms to promote UV-induced apoptosis (Liu *et al.*, 2006). Whether c-Jun inhibits UV-induced cell death under this condition has yet to be determined.

Toward cell signaling-based therapy of human diseases and cancer

The study on signal transduction over past several decades has generated enormous wealth of knowledge regarding cell signaling circuitry, which governs most, if not all, cellular activities. Deregulation of the cell signaling circuitry has been implicated in many human diseases and certain cancer. This knowledge has greatly influenced and will eventually revolutionize the modern medicine. Today, selectively targeting a signaling component in a specific signaling pathway for the therapeutic purpose is no longer a scientific fiction. Naturally, the key signaling regulators like NF- κ B and JNK1 have become attractive therapeutic targets.

Paradoxically, the key signaling regulators like NF- κ B and JNK1 are often essential or critical for normal physiological functions of many tissues and organs in human body. A major concern in the development of cell signaling-based therapy is how to specifically target the key signaling regulators like NF- κ B and JNK1 in a particular tissue or organ without affecting normal functions of other tissues or organs, or even other type of cells in the same organ considering the intra- and intercellular crosstalk among individual signaling pathways. For instance, both biochemical and genetic studies have clearly demonstrated that IKK β , via NF- κ B, protects cells from TNF- α or other death insults (Karin and Lin, 2002). Therefore, it was expected that inhibition of IKK β in hepatocytes may suppress tumorigenesis (Maeda *et al.*, 2005). Surprisingly, it was found that genetic inactivation of IKK β in hepatocytes in mouse liver promoted hepatocarcinogenesis (Maeda *et al.*, 2005). This is because the dying hepatocytes activate neighboring Kupffer cells, which release proinflammatory cytokines to induce a compensatory proliferation in surviving hepatocytes (Maeda *et al.*, 2005). In addition, it should be realized that the same key signaling regulator may elicit different or even opposite biological functions in response to distinct signals. For example, although activation of JNK1 is required for proinflammatory cytokine TNF- α -induced apoptosis in hepatocytes and fibroblasts (Tang *et al.*, 2001; Maeda *et al.*, 2005), JNK1 is also essential for hematopoietic cytokine IL-3-induced cell survival in B lymphocytes (Yu *et al.*, 2004). Therefore, simply targeting a key signaling regulator may not be the best choice in cell signaling-based therapy.

Understanding how the cell signaling circuitry is wired in response to distinct extracellular stimuli may hold the key to design cell signaling-based therapy. The biological functions of key signaling molecules like NF- κ B and JNK1 depend on the activities of other signaling

molecules in the cell signaling circuitry. Although transient JNK1 contributes to TNF- α -induced cell proliferation and inflammation, JNK1 activation becomes prolonged in the absence of NF- κ B activation and promotes TNF- α -induced cell death (Tang *et al.*, 2001, 2002). The biological outcome of the interplay between NF- κ B and JNK1 also depends on the nature of extracellular stimulus (Lin, 2003). NF- κ B negatively regulates TNF- α -induced JNK1 activation for cell survival, but positively regulates UV-induced JNK1 activation for cell death (Tang *et al.*, 2001; Liu *et al.*, 2006). Consequently, inhibition of NF- κ B suppresses certain types of cancer, including certain leukemia and lymphomas (Lin and Karin, 2003), but may promote γ -radiation-induced cancer, as the pre-existing nuclear RelA/NF- κ B is required for γ -radiation-induced cell death (Liu J and Lin A, unpublished data). Wiring the cell signaling circuitry may provide alternative therapeutic targets. Despite NF- κ B activation, TNF- α still induces prolonged JNK1 activation and cell death in breast carcinoma MCF-7 cells (Tang *et al.*, 2002). This suggests that NF- κ B activation is uncoupled in these cells from its ability to prevent prolonged JNK1 activation (Tang *et al.*, 2002). Thus, targeting the NF- κ B-induced JNK1 inhibitor(s) may lead to the prolonged JNK1 activation and cell death without inhibiting NF- κ B. Although it is still a long road to go before the clinicians can prescribe such integrated cell signaling-based medicine to patients who suffer from various diseases and cancer, understanding the wiring of the cell signaling circuitry in response to distinct extracellular stimuli should greatly accelerate this process.

Acknowledgements

This work is partially supported by grants from the National Institutes of Health. We apologize to the authors for not being able to directly cite their work due to the space constraints.

References

- Adler V, Franklin CC, Kraft AS. (1992). Phorbol esters stimulate the phosphorylation of c-Jun but not v-Jun: regulation by the N-terminal delta domain. *Proc Natl Acad Sci USA* **89**: 5341–5345.
- Amanullah A, Azam N, Balliet A, Hollander C, Hoffman B, Fornace A *et al.* (2003). Cell signalling: cell survival and a Gadd45-factor deficiency. *Nature* **424**: 741.
- Antonyak MA, Kenyon LC, Godwin AK, James DC, Emler DR, Okamoto I *et al.* (2002). Elevated JNK activation contributes to the pathogenesis of human brain tumors. *Oncogene* **21**: 5038–5046.
- Baeuerle PA, Baltimore D. (1996). NF- κ B: ten years after. *Cell* **87**: 13–20.
- Baldwin Jr AS. (1996). The NF- κ B and I κ B proteins: new discoveries and insights. *Annu Rev Immunol* **14**: 649–683.
- Barnes PJ, Karin M. (1997). NF- κ B: a pivotal transcription factor in chronic inflammatory diseases. *N Engl J Med* **336**: 1066–1071.
- Beg AA, Baltimore D. (1996). An essential role for NF- κ B in preventing TNF- α -induced cell death. *Science* **274**: 782–784.
- Bender KM, Gottlicher S, Whiteside H, Rahmsdorf J, Herrlich P. (1998). Sequential DNA damage-independent and -dependent activation of NF- κ B by UV. *EMBO J* **17**: 5170–5181.
- Binetruy B, Smeal T, Karin M. (1991). Ha-Ras augments c-Jun activity and stimulates phosphorylation of its activation domain. *Nature* **351**: 122–127.
- Campbell KJ, Rocha S, Perkins ND. (2004). Active repression of antiapoptotic gene expression by RelA(p65) NF- κ B. *Mol Cell* **13**: 853–865.
- Chang L, Karin M. (2001). Mammalian MAP kinase signalling cascades. *Nature* **410**: 37–40.
- Chang L, Kamata H, Solinas G, Luo JL, Maeda S, Venuprasad K *et al.* (2006). The E3 ubiquitin ligase itch couples JNK activation to TNF α -induced cell death by inducing c-FLIP(L) turnover. *Cell* **124**: 601–613.
- Courtois G, Gilmore TD. (2006). Mutations in the NF- κ B signaling pathway: implications for human disease. *Oncogene* **25**: 6831–6843.
- Cripe LD, Gelfanov VM, Smith EA, Spigel DR, Phillips CA, Gabig TG *et al.* (2002). Role for c-jun N-terminal kinase in

- treatment-refractory acute myeloid leukemia (AML): signaling to multidrug-efflux and hyperproliferation. *Leukemia* **16**: 799–812.
- Davis RJ. (2000). Signal transduction by the JNK group of MAP kinases. *Cell* **103**: 239–252.
- De Smaele E, Zazzeroni F, Papa S, Nguyen DU, Jin R, Jones J *et al.* (2001). Induction of gadd45beta by NF-kappa B downregulates pro-apoptotic JNK signalling. *Nature* **414**: 308–313.
- Deng Y, Ren X, Yang L, Lin Y, Wu X. (2003). A JNK-dependent pathway is required for TNFalpha-induced apoptosis. *Cell* **115**: 61–70.
- Derijard B, Hibi M, Wu IH, Barrett T, Su B, Deng T *et al.* (1994). JNK1: a protein kinase stimulated by UV light and Ha-Ras that binds and phosphorylates the c-Jun activation domain. *Cell* **76**: 1025–1037.
- Derijard B, Raingeaud J, Barrett T, Wu IH, Han J, Ulevitch RJ *et al.* (1995). Independent human MAP-kinase signal transduction pathways defined by MEK and MKK isoforms. *Science* **267**: 682–685.
- Ducut Sigala JL, Bottero V, Young DB, Shevchenko A, Mercurio F, Inder M *et al.* (2004). Activation of transcription factor NF-kappa B requires ELKS, an Ikappa B kinase regulatory subunit. *Science* **304**: 1963–1967.
- Enari M, Sakahira H, Yokoyama H, Okawa K, Iwamatsu A, Nagata S. (1998). A caspase-activated DNase that degrades DNA during apoptosis, and its inhibitor ICAD. *Nature* **391**: 43–50.
- Gao M, Labuda T, Xia Y, Gallagher E, Fang D, Liu YC *et al.* (2004). Jun turnover is controlled through JNK-dependent phosphorylation of the E3 ligase Itch. *Science* **306**: 271–275.
- Ghosh S, Karin M. (2002). Missing pieces in the NF- κ B puzzle. *Cell* **109**: S81–S96.
- Ghosh S, May MJ, Kopp EB. (1998). NF- κ B and Rel proteins: evolutionarily conserved mediators of immune responses. *Annu Rev Immunol* **16**: 225–260.
- He H, Li HL, Lin A, Gottlieb RA. (1999). Activation of the JNK pathway is important for cardiomyocyte death in response to simulated ischemia. *Cell Death Differ* **6**: 987–991.
- Herrlich P, Blattner C, Knebel A, Bender K, Rahmsdorf HJ. (1997). Nuclear and non-nuclear targets of genotoxic agents in the induction of gene expression. Shared principles in yeast, rodents, man and plants. *Biol Chem* **378**: 1217–1229.
- Hibi M, Lin A, Smeal T, Minden A, Karin M. (1993). Identification of an oncoprotein- and UV-responsive protein kinase that binds and potentiates the c-Jun activation domain. *Genes Dev* **7**: 2135–2148.
- Hirosumi J, Tuncman G, Chang L, Görgün CZ, Uysal KT, Maeda K *et al.* (2002). A central role for JNK in obesity and insulin resistance. *Nature* **420**: 333–336.
- Hutchison M, Berman KS, Cobb MH. (1998). Isolation of TAO1, a protein kinase that activates MEKs in stress-activated protein kinase cascade. *J Biol Chem* **273**: 28625–28632.
- Ichijo H, Nishida E, Irie K, ten Dijke P, Saitoh M, Moriguchi T *et al.* (1997). Induction of apoptosis by ASK1 a mammalian MAPKKK that activates SAPK/JNK and p38 signaling pathways. *Science* **275**: 90–94.
- Ito M, Yoshioka K, Akechi M, Yamashita S, Takamatsu N, Sugiyama K *et al.* (1999). JSAP1, a novel jun N-terminal protein kinase (JNK)-binding protein that functions as a Scaffold factor in the JNK signaling pathway. *Mol Cell Biol* **19**: 7539–7548.
- Kamata H, Honda S, Maeda S, Chang L, Hirata H, Karin M. (2005). Reactive oxygen species promote TNF- α -induced death and sustained JNK activation by inhibiting MAP kinase phosphatases. *Cell* **120**: 649–661.
- Kasibhatla S, Brunner T, Genestier L, Echeverri LF, Mahboubi A, Green DR. (1998). DNA damaging agents induce expression of Fas ligand and subsequent apoptosis in T lymphocytes via the activation of NF-kappa B and AP-1. *Mol Cell* **1**: 543–551.
- Karin M. (1995). The regulation of AP-1 activity by mitogen-activated protein kinases. *J Biol Chem* **270**: 16483–16486.
- Karin M. (2006). Nuclear factor-kappa B in cancer development and progression. *Nature* **441**: 431–436.
- Karin M, Ben-Neriah Y. (2000). Phosphorylation meets ubiquitination: the control of NF- κ B activity. *Annu Rev Immunol* **18**: 621–663.
- Karin M, Lin A. (2002). NF- κ B at the crossroad of life and death. *Nat Immunol* **3**: 221–227.
- Kato Jr T, Delhase M, Hoffman A, Karin M. (2003). CK2 is a C-terminal I κ B kinase responsible for NF- κ B activation during the UV response. *Mol Cell* **12**: 829–839.
- Khelifi AF, D'Alcontres MS, Salomoni P. (2005). Daxx is required for stress-induced cell death and JNK activation. *Cell Death Differ* **12**: 724–733.
- Lee SA, Dritschilo A, Jung M. (1998). Impaired ionizing radiation-induced activation of a nuclear signal essential for phosphorylation of c-Jun by dually phosphorylated c-Jun amino-terminal kinases in ataxia telangiectasia fibroblasts. *J Biol Chem* **273**: 32889–32894.
- Lei K, Davis RJ. (2003). JNK phosphorylation of Bim-related members of the Bcl2 family induces Bax-dependent apoptosis. *Proc Natl Acad Sci USA* **100**: 2432–2437.
- Li N, Karin M. (1998). Ionizing radiation and short wavelength UV activate NF- κ B through two distinct mechanisms. *Proc Natl Acad Sci USA* **95**: 13012–13017.
- Li Q, Van Antwerp D, Mercurio F, Lee KF, Verma IM. (1999a). Severe liver degeneration in mice lacking the I κ B kinase 2 gene. *Science* **284**: 321–325.
- Li ZW, Chu W, Hu Y, Delhase M, Deerinck T, Ellisman M *et al.* (1999b). The IKK β subunit of I κ B kinase (IKK) is essential for nuclear factor κ B activation and prevention of apoptosis. *J Exp Med* **189**: 1839–1845.
- Lin A. (2003). Activation of the JNK signaling pathway: breaking the brake on apoptosis. *BioEssays* **25**: 17–24.
- Lin A (ed). (2006). *The JNK Signaling Pathway*. Landes Bioscience: TX.
- Lin A, Karin M. (2003). NF- κ B in cancer: a marked target. *Semin Cancer Biol* **13**: 107–114.
- Lin A, Minden A, Martinetto H, Claret FX, Lange-Carter C, Mercurio F *et al.* (1995). Identification of a dual specificity kinase that activates the Jun kinases and p38-Mpk2. *Science* **268**: 286–290.
- Liu J, Minemoto Y, Lin A. (2004). JNK1, but not JNK2, is essential for TNF- α -induced c-Jun kinase activation and apoptosis. *Mol Cell Biol* **24**: 10844–10856.
- Liu J, Yang D, Minemoto Y, Leitges M, Rosner MR, Lin A. (2006). NF- κ B via PKC δ promotes JNK activation by UV. *Molecular Cell* **21**: 467–480.
- Liu ZG, Hsu H, Goeddel DV, Karin M. (1996). Dissection of TNF receptor 1 effector functions: JNK activation is not linked to apoptosis while NF-kappa B activation prevents cell death. *Cell* **87**: 565–576.
- Lopez-Bergami P, Habelhah H, Bhoumik A, Zhang W, Wang LH, Ronai Z. (2006). RACK1 mediates activation of JNK by protein kinase C. *Mol Cell* **19**: 309–320.
- Lu C, Zhu F, Cho YY, Tang F, Zykova T, Ma WY *et al.* (2006). Cell apoptosis: requirement of H2AX in DNA ladder formation, but not for the activation of caspase-3. *Mol Cell* **23**: 121–132.
- Lu X, Nemoto S, Lin A. (1997). Identification of c-Jun NH2-terminal protein kinase (JNK)-activating kinase 2 as

- an activator of JNK but not p38. *J Biol Chem* **272**: 24751–24754.
- Madrid LV, Wang CY, Guttridge DC, Schottelius AJG, Albert S, Baldwin Jr AS *et al.* (2000). Akt suppresses apoptosis by stimulating the transactivation potential of the RelA/p65 subunit of NF- κ B. *Mol Cell Biol* **20**: 1626–1638.
- Maeda S, Chang L, Li Z-W, Luo J-L, Leffert H, Karin M. (2003). IKK β is required for prevention of apoptosis mediated by cell bound but not by circulating TNF α . *Immunity* **19**: 725–737.
- Maeda S, Kamata H, Luo JL, Leffert H, Karin M. (2005). IKK β couples hepatocyte death to cytokine-driven compensatory proliferation that promotes chemical hepatocarcinogenesis. *Cell* **121**: 977–990.
- McDonald PH, Chow CW, Miller WE, Laporte SA, Field ME *et al.* (2000). Beta-arrestin 2: a receptor-regulated MAPK scaffold for the activation of JNK3. *Science* **290**: 1574–1577.
- Micheau O, Tschopp J. (2003). Induction of TNF receptor I-mediated apoptosis via two sequential signaling complexes. *Cell* **114**: 181–190.
- Natoli G, Costanzo A, Ianni A, Templeton DJ, Woodgett JR, Balsano C *et al.* (1997). Activation of SAPK/JNK by TNF receptor 1 through a noncytotoxic TRAF2-dependent pathway. *Science* **275**: 200–203.
- Nemoto S, Sheng Z, Lin A. (1998). Opposing effects of Jun kinase and p38 mitogen-activated protein kinases on cardiomyocyte hypertrophy. *Mol Cell Biol* **18**: 3518–3526.
- Nijhawan D, Fang M, Traer E, Zhong Q, Gao W, Du F *et al.* (2003). Elimination of Mcl-1 is required for the initiation of apoptosis following ultraviolet irradiation. *Genes Dev* **17**: 1475–1486.
- Okazawa H, Estus S. (2002). The JNK/c-Jun cascade and Alzheimer's disease. *Am J Alzheimers Dis Other Dement* **17**: 79–88.
- Pando MP, Verma IM. (2000). Signal-dependent and -independent degradation of free and NF- κ B-bound I κ B α . *J Biol Chem* **275**: 21278–21286.
- Papa S, Zazzeroni F, Bubici C, Jayawardena S, Alvarez K, Matsuda S *et al.* (2004). Gadd45 beta mediates the NF-kappa B suppression of JNK signalling by targeting MKK7/JNK2. *Nat Cell Biol* **6**: 146–153.
- Park JM, Brady H, Ruocco MG, Sun H, Williams D, Lee SJ *et al.* (2004). Targeting of TAK1 by the NF-kappa B protein Relish regulates the JNK-mediated immune response in *Drosophila*. *Genes Dev* **18**: 584–594.
- Pham CG, Bubici C, Zazzeroni F, Papa S, Jones J, Alvarez K *et al.* (2004). Ferritin heavy chain upregulation by NF-kappa B inhibits TNF α -induced apoptosis by suppressing reactive oxygen species. *Cell* **119**: 529–542.
- Pulverer BJ, Kyriakis JM, Avruch J, Nikolakaki E, Woodgett JR. (1991). Phosphorylation of c-jun mediated by MAP kinases. *Nature* **353**: 670–674.
- Purcell NH, Yu C, He D, Xiang J, Paran N, DiDonato JA *et al.* (2001). Activation of NF- κ B by hepatitis B virus X protein through an I κ B kinase-independent mechanism. *Am J Physiol* **280**: G669–G677.
- Reimold AM. (2002). TNF α as therapeutic target: new drugs, more applications. *Curr Drug Targets Inflamm Allergy* **1**: 377–392.
- Rosette C, Karin M. (1996). Ultraviolet light and osmotic stress: activation of the JNK cascade through multiple growth factor and cytokine receptors. *Science* **274**: 1194–1197.
- Sabapathy K, Hochedlinger K, Nam SY, Bauer A, Karin M, Wagner EF. (2004). Distinct roles for JNK1 and JNK2 in regulating JNK activity and c-Jun-dependent cell proliferation. *Mol Cell* **15**: 834–844.
- Sadoshima J, Montagne O, Wang Q, Yang G, Warden J, Liu J *et al.* (2002). The MEKK1-JNK pathway plays a protective role in pressure overload but does not mediate cardiac hypertrophy. *J Clin Invest* **110**: 271–279.
- Sakon S, Xue X, Takekawa M, Sasazuki T, Okazaki T, Kojima Y *et al.* (2003). NF-kappa B inhibits TNF-induced accumulation of ROS that mediate prolonged MAPK activation and necrotic cell death. *EMBO J* **22**: 3898–3909.
- Salmeron A, Ahmad TB, Carlile GW, Pappin D, Narsimhan RP, Ley SC. (1996). Activation of MEK-1 and SEK-1 by Tpl-2 proto-oncoprotein, a novel MAP kinase kinase. *EMBO J* **15**: 817–826.
- Sanchez I, Hughes RT, Mayer BJ, Yee K, Woodgett JR, Avruch J *et al.* (1994). Role of SAPK/ERK kinase-1 in the stress-activated pathway regulating transcription factor c-Jun. *Nature* **372**: 794–798.
- Shaulian E, Schreiber M, Piu F, Beeche M, Wagner EF, Karin M. (2000). The mammalian UV response: c-Jun induction is required for exit from p53-imposed growth arrest. *Cell* **103**: 897–907.
- She QB, Chen N, Bode AM, Flavell RA, Dong Z. (2002). Deficiency of c-Jun-NH(2)-terminal kinase-1 in mice enhances skin tumor development by 12-O-tetradecanoylphorbol-13-acetate. *Cancer Res* **62**: 1343–1348.
- Shibuya H, Yamaguchi K, Shirakabe K, Toneyawa A, Gotoh Y, Ueno N *et al.* (1996). TAB1: an activator of the TAK1 MAPKKK in TGF-beta signal transduction. *Science* **272**: 1179–1182.
- Siebenlist U, Franzoso G, Brown K. (1994). Structure, regulation and function of NF- κ B. *Annu Rev Cell Biol* **10**: 405–455.
- Smeal T, Binetruy B, Mercola DA, Birrer M, Karin M. (1991). Oncogenic and transcriptional cooperation with Ha-Ras requires phosphorylation of c-Jun on serines 63 and 73. *Nature* **354**: 494–496.
- Tanaka M, Fuentes ME, Yamaguchi K, Durnin MH, Dalrymple SA, Hardy KL *et al.* (1999). Embryonic lethality, liver degeneration, and impaired NF- κ B activation in IKK β -deficient mice. *Immunity* **10**: 421–429.
- Tang F, Tang G, Xiang J, Dai Q, Rosner MR, Lin A. (2002). Absence of NF- κ B-mediated inhibition of c-Jun N-terminal kinase activation contributes to tumor necrosis factor alpha-induced apoptosis. *Mol Cell Biol* **22**: 8571–8579.
- Tang G, Minemoto Y, Dibling B, Purcell NH, Li Z, Karin M *et al.* (2001). Inhibition of JNK activation through NF- κ B target genes. *Nature* **414**: 313–317.
- Thanos D, Maniatis T. (1995). NF- κ B: a lesson in family values. *Cell* **80**: 529–532.
- Tibbles LA, Ing YL, Kiefer F, Chan J, Iscove N, Woodgett JR *et al.* (1996). MLK-3 activates the SAPK/JNK and p38/RK pathways via SEK1 and MKK3/6. *EMBO J* **15**: 7026–7035.
- Tournier C, Hess P, Yang DD, Xu J, Turner TK, Nimnual A *et al.* (2000). Requirement of JNK for stress-induced activation of the cytochrome c-mediated death pathway. *Science* **288**: 870–874.
- Tournier C, Whitmarsh AJ, Cavanagh J, Barrett T, Davis RJ. (1997). Mitogen-activated protein kinase kinase 7 is an activator of the c-Jun NH2-terminal kinase. *Proc Natl Acad Sci USA* **94**: 7337–7342.
- Uehara T, Bennett B, Sakata S, Satoh Y, Bilter G, Westwick J *et al.* (2005). JNK mediates hepatic ischemia reperfusion injury. *J Hepatol* **42**: 850–859.

- Van Antwerp DJ, Martin SJ, Kafri T, Green DR, Verma IM. (1996). Suppression of TNF- α -induced apoptosis by NF- κ B. *Science* **274**: 787–789.
- Verheij M, Bose R, Lin XH, Yao B, Jarvis WD, Grants S *et al.* (1996). Requirement for ceramide-initiated SAPK/JNK signalling in stress-induced apoptosis. *Nature* **380**: 75–79.
- Verma IM, Stevenson JK, Schwarz EM, Van Antwerp D, Miyamoto S. (1995). Rel/NF- κ B/I κ B family: intimate tales of association and dissociation. *Genes Dev* **9**: 2723–2735.
- Whitmarsh AJ, Cavanagh J, Tournier C, Yasuda J, Davis RJ. (1998). A mammalian scaffold complex that selectively mediates MAP kinase activation. *Science* **281**: 1671–1674.
- Wisdom R, Johnson RS, Moore C. (1999). c-Jun regulates cell cycle progression and apoptosis by distinct mechanisms. *EMBO J* **18**: 188–197.
- Wu CJ, Conze DB, Li X, Ying SX, Hanover JA, Ashwell JD. (2005). TNF- α induced c-IAP1/TRAF2 complex translocation to a Ubc6-containing compartment and TRAF2 ubiquitination. *EMBO J* **24**: 1886–1898.
- Xia XG, Harding T, Weller M, Bieneman A, Uney JB, Schulz JB. (2001). Gene transfer of the JNK interacting protein-1 protects dopaminergic neurons in the MPTP model of Parkinson's disease. *Proc Natl Acad Sci USA* **98**: 10433–10438.
- Xia Y, Makris C, Su B, Li E, Yang J, Nemerow GR *et al.* (2000). MEK kinase 1 is critically required for c-Jun N-terminal kinase activation by proinflammatory stimuli and growth factor-induced cell migration. *Proc Natl Acad Sci USA* **97**: 5243–5248.
- Yu C, Minemoto Y, Zhang J, Liu J, Tang F, Bui TN *et al.* (2004). JNK suppresses apoptosis via phosphorylation of the proapoptotic Bcl-2 family protein BAD. *Mol Cell* **13**: 329–340.
- Yujiri T, Sather S, Fanger GR, Johnson GL. (1998). Role of MEKK1 in cell survival and activation of JNK and ERK pathways defined by targeted gene disruption. *Science* **282**: 1911–1914.
- Zhang J, Bui T, Xiang J, Lin A. (2006). cAMP Inhibits p38 Activation via CREB-induced Dynein Light Chain. *Mol Cell Biol* **26**: 1223–1234.
- Zheng C, Xiang J, Hunter T, Lin A. (1999). The JNKK2-JNK1 fusion protein acts as a constitutively active c-Jun kinase that stimulates c-Jun transcription activity. *J Biol Chem* **274**: 28966–28971.